Strip R-CNN: Large Strip Convolution for Remote Sensing Object Detection

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Project page: https://github.com/HVision-NKU/Strip-R-CNN

Abstract

While witnessed with rapid development, remote sensing object detection remains challenging for detecting high aspect ratio objects. This paper shows that large strip convolutions are good feature representation learners for remote sensing object detection and can detect objects of various aspect ratios well. Based on large strip convolutions, we build a new network architecture called Strip R-CNN, which is simple, efficient, and powerful. Unlike recent remote sensing object detectors that leverage large-kernel convolutions with square shapes, our Strip R-CNN takes advantage of sequential orthogonal large strip convolutions in our backbone network ***StripNet*** *to capture spatial information. In addition, we improve the localization capability of remote-sensing object detectors by decoupling the detection heads and equipping the localization branch with strip convolutions in our* ***strip head****. Extensive experiments on several benchmarks, for example DOTA, FAIR1M, HRSC2016, and DIOR, show that our Strip R-CNN can greatly improve previous work. In particular, our 30M model achieves 82.75% mAP on DOTA-v1.0, setting a new state-of-the-art record. Our code will be made publicly available.*

1. Introduction

Remote sensing object detection has gained significant attention in recent years due to its application in aerial images captured by drones and satellites [30, 31, 34, 35, 41, 44, 57, 78]. A popular pipeline is built on the basis of rotated boxes to cover objects of interest. Due to boundary discontinuity and square-like problems [67, 71, 72] and the urgent need to capture long-range information [1, 32], many research

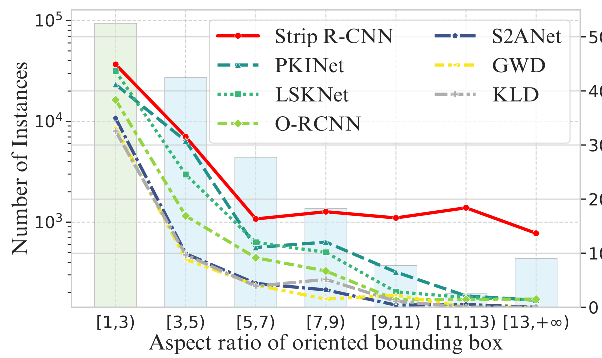

Figure 1. Statistics of the DOTA dataset [62] and the detection performance of several recent state-of-the-art detectors. We can see that slender objects (aspect ratio > 3) occupy a non-negligible proportion and detection performance of previous state-of-the-art models declines as aspect ratio increases.

breakthroughs have been made to develop stronger rotated object detectors, including object representations [29, 63, 64, 66], IoU-simulated loss functions [67, 71–73], and foundation models [1, 32, 51], etc.

Despite the great progress made by previous work, successfully detecting high aspect ratio objects, which are prevalent in remote sensing object detection, is still a challenging problem. To demonstrate this, in Fig. 1, we calculate the statistics of the widely used DOTA dataset [62], which shows that slender objects are quite common in remote sensing scenarios and usually occupy a large proportion of the data. We also experimentally found that existing object detection methods [1, 19, 32, 64, 71, 72] often struggle with slender objects, where detection performance decreases as the aspect ratio of objects increases.

We argue that the difficulties in detecting these objects arise from two primary challenges. First, *high aspect ratio*

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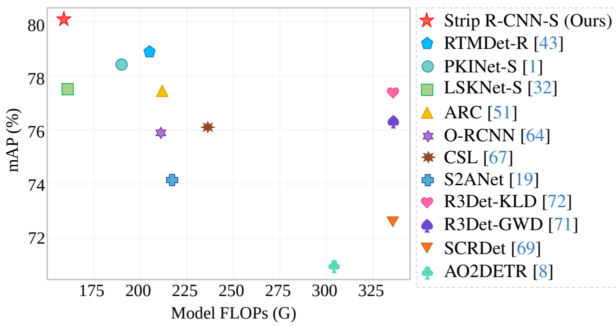


Figure 2. A comprehensive comparison of detection performance on the DOTA dataset of various remote sensing object detectors.

objects contain rich feature information along one spatial dimension, while exhibiting relatively sparse feature in the other. Traditional detectors based on convolutional neural networks mostly extract input feature maps within square windows. This design greatly restricts their ability to effectively capture the anisotropic context, which can be commonly found in remote sensing images, leading to the inclusion of irrelevant information from surrounding areas. Second, *high aspect ratio objects pose considerable challenges in regression tasks due to their unique geometric properties.* In remote sensing object detection, unlike general object detection, an additional angle regression is required. For high aspect ratio objects, even a small error in angle estimation can lead to significant deviations from the ground truth.

To date, there are few works considering how to deal with challenging high aspect ratio objects. A generic approach widely used in previous methods is to enlarge the receptive field of models with large-kernel convolutions [3, 11, 12, 37]. A typical example should be LSKNet [32], which introduces large-kernel convolutions with a spatial selection mechanism to capture long-range contextual information. PKINet [1] further extends LSKNet with a parallel large square conv structure to enhance the performance for large variation of object scales. However, the parallel paradigm of using multiple large-kernel convolutions exacerbates the computational burden, and the complicated block design further restricts model efficiency. For the high variation of the object aspect ratio, how to efficiently use large-kernel convolutions is still an open question.

In this paper, we propose Strip R-CNN, which can efficiently combine the advantage of the square convolution and the strip convolution with less feature redundancy. Our design principles are two-fold. First, the new network architecture should be simple and efficient. Second, it should be good at handling objects of different aspect ratios even when they are high. Given the characteristics of the objects in remote sensing images, we propose using orthogonal large strip convolutions as the main spatial filters, which comprise the core component of our **StripNet** backbone, called strip module. As such, our network is quite simple but can generalize well to even objects of especially high aspect ratios as shown in Fig. 1. Furthermore, to conquer the second challenge mentioned above, we decouple the detection heads used in previous remote sensing object detections and strengthen the localization branch with strip convolutions in our **strip head**. We found that this can not only assist in improving the localization capability of our method but also help more accurately regress the box angles.

To our knowledge, Strip R-CNN is the first work to explore how to take advantage of large strip convolutions for remote sensing object detection. Despite simplicity and the lightweight nature, our Strip R-CNN achieves state-of-the-art performance on the standard DOTA benchmark without bells and whistles, as shown in Fig. 2. Notably, our model Strip R-CNN-S with only 30M learnable parameters achieves the best results on DOTA leaderboard with 82.75% mAP. We also conduct extensive experiments on several other remote sensing datasets, including FAIR1M, HRSC2016, and DIOR, and show the superiority of our Strip R-CNN over other methods. We hope that our design principles could provide new research insights for the remote sensing imagery community.

2. Related Work

Remote Sensing Object Detection. In remote sensing scenarios, where objects are arbitrarily oriented, rotated bounding boxes are generally adopted for object representation. Early representations of rotated bounding boxes use five parameters (x, y, w, h, θ) [10, 39, 69, 77]. However, due to the periodicity of the angle, training models based on this representation often face boundary discontinuity problems in regression [67, 71, 73]. To address this issue, several approaches propose improved representations [15, 29, 59, 63, 64, 66, 70, 74] for rotated bounding boxes. For instance, Oriented R-CNN [64] replaces the angle with midpoint offsets, leading to a six-parameter representation $(x, y, w, h, \triangle\alpha, \triangle\beta)$, which significantly enhances detector performance. COBB [63] introduces a continuous representation of rotated boxes with nine parameters based on the aspect ratio of the minimum enclosing rectangle to the rotated bounding box areas. There are also approaches focusing on mitigating the discontinuity problem in boundary regression through loss functions [7, 24, 52, 71, 72]. For example, GWD [71] and KLD [72] convert rotated bounding boxes into 2D Gaussian distributions and use Gaussian Wasserstein Distance and Kullback-Leibler Divergence as the loss functions. KFIoU [73] uses Gaussian modeling and Kalman filtering and propose the SKewIoU Loss.

Despite the great progress made by previous work, successfully detecting high aspect ratio objects, which are prevalent in remote sensing object detection, is still a chal-

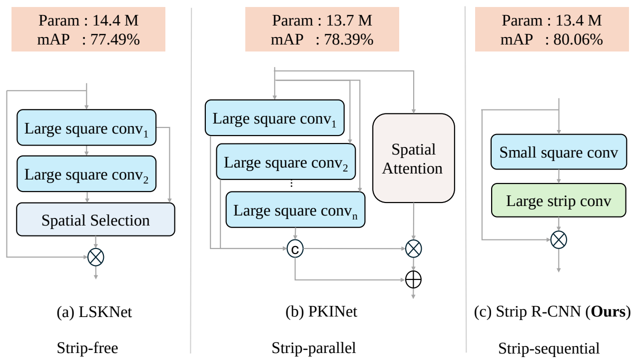


Figure 3. Structural comparison between our proposed strip module and other representative methods using large-kernel convolutions, including LSKNet [32] and PKINet [1].

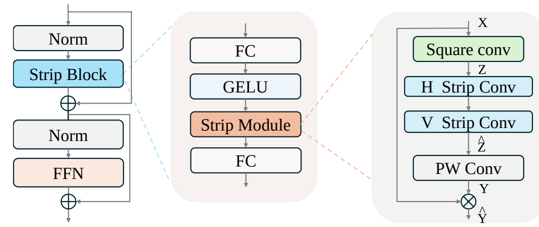


Figure 4. Structure of our basic block of StripNet backbone.

Table 1. Variants of StripNet backbone. C_i: feature channel number; D_i: number of strip blocks in each stage i.

Model	$\{C_1, C_2, C_3, C_4\}$	$\{D_1, D_2, D_3, D_4\}$	#P	FLOPs
⋆ StripNet-T	{32, 64, 160, 256}	{3, 3, 5, 2}	3.8M	18.2G
⋆ StripNet-S	{64, 128, 320, 512}	{2, 2, 4, 2}	13.3M	52.3G

lenging problem. Our method carefully considers the difficulties posed by these objects and take advantage of large strip convolutions to make our network generalize well to the challenging slender objects, which to our knowledge has not been explored before in this research field.

Large Kernel Networks for Remote Sensing Object Detection. Convolution with large kernel has been an emerging and promising solution to remote sensing object detection, which has been validated to have highly competitive performance against the Transformer-based methods in image classification and segmentation [3, 11, 12, 16, 17, 25, 26, 37, 75]. In remote sensing object detection, some approaches put efforts on employing large-kernel convolutions to get long-range contextual information [1, 33]. For example, LSKNet [33] utilizes large kernel convolutions and a selection mechanism to model the contextual information needed for different object categories. PKINet [1] arranges multiple large-kernel convolutions in parallel to extract dense texture features across diverse receptive fields, and introduces a context anchor attention mechanism to capture relationships between distant pixels. However, the parallel paradigm of leveraging multiple large-kernel convolutions exacerbates the computational burden, and the complicated block design makes the model not efficient. Regarding the high variation of the object aspect ratio, how to efficiently make use of large-kernel convolutions is still an open question. To our knowledge, Strip R-CNN is the first work to explore how to take advantage of large strip convolutions for remote sensing object detection.

3. Strip R-CNN

In this section, we describe the architecture of the proposed Strip R-CNN in detail. Our goal is to advance remote sensing object detectors with large strip convolutions so that the resulting model can perform well on objects of different aspect ratios. This is different from previous work that emphasizes the importance of convolutions with large square kernels, as shown in Fig. 3.

3.1. Overall Architecture

Based on the O-RCNN framework [64], our Strip R-CNN replace the backbone and detection head with our **StripNet backbone** and **strip head**, respectively. Specifically, the backbone is mainly composed of basic blocks proposed as illustrated in Fig. 4, which consists of two residual sub-blocks: the strip sub-block and the feed-forward network sub-block. The strip sub-block is built upon a small-kernel standard convolution and two convolutions with large strip-shaped kernels to capture robust features for objects of different aspect ratios. For the feed-forward network sub-block, we simply follow LSKNet [32], which is used to facilitate channel mixing and feature refinement. The detailed configurations of different variants of our StripNet backbone are shown in Tab. 1. For the stem layers, we keep them the same to LSKNet [32]. For the detection head, we decouple the localization branch from the original O-RCNN head and enhance it with the proposed strip module, resulting in our strip head, which will be presented in Sec. 3.3.

3.2. Strip Module

As discussed in Sec. 1, large square kernel convolutions provide essential long-range contextual information for remote sensing applications. PKINet [1] introduces parallel large square kernel convolutions and spatial attention mechanism to extract spatial information. However, the parallel paradigm of leveraging multiple large-kernel convolutions exacerbates the computational burden, and the complicated block design makes the model not efficient. Our objective is to efficiently extract essential features for objects of varying aspect ratios. The outcome is a sequential paradigm that efficiently combines the advantages of both standard and strip convolutions without requiring additional information fusion module. In what follows, we provide a detailed de-

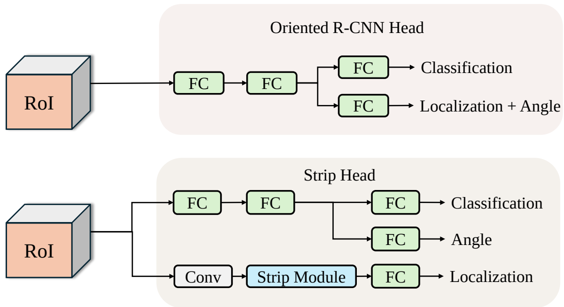


Figure 5. Structural comparison of Oriented R-CNN head and strip head. In our strip head, the classification and angle prediction heads share two fully connected layer, while the localization head incorporates our strip module.

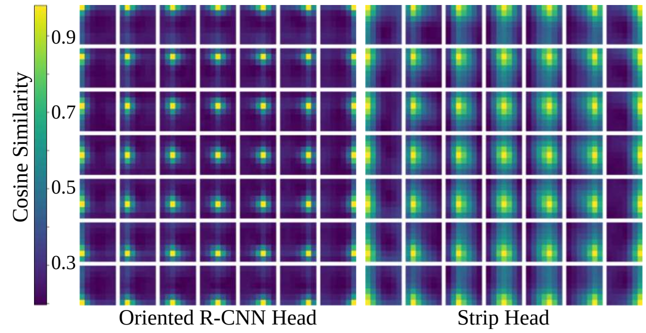


Figure 6. Spatial correlation map comparison of the Oriented R-CNN head and our strip head. Our strip head has significantly more spatial correlations in the output feature maps.

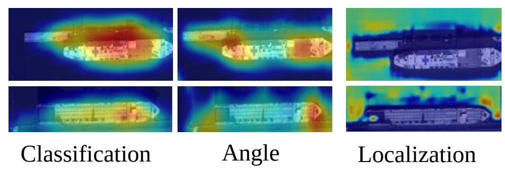


Figure 7. Spatial sensitivity heatmap comparison for classification, localization and angle prediction in the output feature maps.

scription of the core of our basic block: **strip module**.

Given an input tensor $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ with C channels, a depthwise convolution with square kernels $\mathbf{K} \in \mathbb{R}^{C\times k_H\times k_W}$ is first applied to extract local contextual features, yielding $\mathbf{Z}$, where $H \times W$ and $k_H \times k_W$ are the feature size and kernel size, respectively. In practical use, we set $k_H \times k_W$ to 5×5. After the initial depthwise convolution, we use two sequential depthwise convolution of large strip-shaped kernels to better capture objects of high aspect ratios. The output is denoted as $\hat{\mathbf{Z}}$. Unlike standard convolutions that extract features from a square region for each time, large strip convolutions allow the network to focus more on features along either the horizontal or the vertical axis. The combined use of horizontal and vertical large strip convolutions enables the network to collect directional features across both spatial axes, enhancing the representations of elongated or narrow structures in spatial dimension.

To further enhance the interaction of the features across the channel dimension, a simple point-wise convolution is applied to transform $\hat{\mathbf{Z}}$ to $\mathbf{Y}$. In this way, each position of the resulting feature map $\mathbf{Y}$ encodes both horizontal and vertical features across a wide spatial area. Finally, following [17, 26], we regard the feature map $\mathbf{Y}$ as attention weights to reweigh the input X, which can be formulated as

$$\hat{\mathbf{Y}} = \mathbf{X} \cdot \mathbf{Y}, \tag{1}$$

where '$\cdot$' denotes the element-wise multiplication operation. Fig. 4 provides a diagrammatic illustration of the proposed strip module. In our experiments, we will discuss how to choose the kernel size of the strip convolutions.

It is important to emphasize that our backbone network StripNet is much simpler than previous remote sensing object detectors using large-kernel convolutions as shown in Fig. 3. We do not utilize any spatial or channel attention mechanisms in our basic block design nor compound fusion operations with different types of large-kernel convolutions. This makes our StripNet quite simple but has great performance on different remote sensing detection benchmarks. Moreover, we found that there is no significant difference in applying horizontal strip convolutions before vertical ones or vice versa. Both approaches are effective.

3.3. Detection Head with Strip Convolutions

In localization tasks, models should be sensitive to transformations, as the accuracy of localization depends on the positions of input objects. Previous strong rotated object detectors [1, 32] adopt the Oriented R-CNN framework [64], whose detection head shares the same linear layers for classification and localization, as shown in Fig. 5. However, linear layers have limited spatial correlation as demonstrated in [61], making them transformation-insensitive and unsuitable for precise localization.

A better solution might be to decouple the classification and localization tasks and using small kernel convolutions in the localization branch as suggested in [61]. However, our analysis of spatial correlation maps shows that small kernel convolutions capture only short-range spatial correlations, as illustrated on the left of Fig. 6. These short-range correlations are inadequate for accurately localizing slender objects, which require long-range dependencies. To effectively localize objects of varying aspect ratios, we argue that the localization head should be able to capture long-range dependencies, similar to those handled by the backbone network. Large strip convolutions, which capture both horizontal and vertical features across a broad spatial area, provide the extended spatial correlations necessary for better localization. Therefore, we propose using large strip convolutions in the localization branch to enhance the detector's localization capabilities.

Furthermore, predicting the parameters x, y, w, h, and θ together may lead to the issue of coupled features [76]. To address this, we decouple the prediction of θ from the other parameters (x, y, w, h). Building on this decoupling approach, we investigate the spatial sensitivity regions of the classification and angle predictions in the output feature maps. The spatial sensitivity heat map is shown in Fig. 7, where the first column represents the classification sensitivity, and the second shows the angle sensitivity. Our observation reveals that the spatial sensitivity for classification is concentrated in the central areas of the objects, while the sensitivity for angle is primarily focused near the object borders. The sensitive regions for classification and angle predictions share some overlap and exhibit some complementary characteristics. Based on this, we propose the use of shared fully connected layers for both classification and angle predictions. The structure of the strip head is shown at the bottom of Fig. 5. Detailed settings are described below.

Classification head: The classification head simply utilizes two fully connected layers with an output dimension of 1024. Using fully connected layers in the classification head has been shown to be a good choice, as demonstrated in Double-Head RCNN [61], so we keep it unchanged.

Localization head: The localization head begins with a standard 3×3 convolution to extract local features. Then we add a strip module, followed by a fully connected layer to collect long-range spatial dependencies.

Angle head: For angle prediction, we adopt three fully connected layers to estimate angular information, which we found works well. Note that the first two fully connected layers share parameters with the classification head.

It is worth noting that our detection head incorporates the proposed strip module. This design largely improves the detection capability compared to the original Oriented R-CNN head. As shown in Fig. 6, from the visualization of the spatial sensitivity of different methods, we can see that the Oriented R-CNN head has a similar spatial correlation pattern on the output feature maps. However, our strip head has much more spatial correlations in the output feature maps. We will show that this enables our method to be able to better localize objects even of higher aspect ratios.

4. Experiments

4.1. Experiment Setup

Datasets. We conduct extensive experiments on five popular remote sensing object detection datasets.

- **DOTA-v1.0** [62] is a large-scale dataset for remote sensing detection which contains 2,806 images, 188,282 instances, and 15 categories.

Table 2. Comparisons with different backbone models on DOTA-v1.0. Params and FLOPs are computed for backbone only. All backbones are pretrained on ImageNet for 300 epochs. Our StripNet-S achieves higher mAP than previous popular backbones.

Model (Backbone)	**#P↓**	**FLOPs↓**	**FPS**	**mAP (%)**
ResNet-50 [21]	23.3M	86.1G	21.8	75.87
LSKNet-S [32]	14.4M	54.4G	20.7	77.49
PKINet-S [1]	13.7M	70.2G	12.0	78.39
⋆StripNet-S	13.3M	52.3G	17.7	80.06

- **DOTA-v1.5** [62] is a more challenging dataset which contains 403,318 instances and 16 categories.
- **FAIR1M-v1.0** [56] is a remote sensing dataset consisting of 15,266 images, 1 million instances, and 37 categories.
- **HRSC2016** [38] is a remote sensing dataset that contains 1061 aerial images and 2,976 instances.
- **DIOR-R** [5] contains 23,463 images and 192,518 instances.

Implementation details. Our training process is divided into two stages: pretraining on ImageNet [9] and fine-tuning on downstream remote sensing datasets. For ablation experiments, we train all the models for 100 epochs. To achieve higher performance, we train the models for 300 epochs for the final results. The number of training epochs for DOTA, DOTAv1.5, HRSC2016, FAIR1M-v1.0, and DIOR-R are set to 12, 12, 36, 12, and 12, respectively, following previous methods. Learning rates are set to 0.0001, 0.0001, 0.0004, 0.0001, and 0.0001, respectively. The input sizes for HRSC2016 and DIOR-R are 800×800, while for the DOTA-v1.0, DOTA-v1.5 and FAIR1M-v1.0 datasets, the input sizes are 1024×1024. During training, we employ the AdamW [42] optimizer with $\beta_1 = 0.9, \beta_2 = 0.999$, and a weight decay of 0.05. All the models are trained on 8 NVIDIA 3090 GPUs with a batch size of 8, and test is conducted on a single NVIDIA 3090 GPU.

4.2. Main Results

We first compare our Strip R-CNN with recent state-of-the-art methods with strong backbones implemented within the Oriented R-CNN [64] framework on the DOTA v1.0 dataset. As shown in Tab. 2, Strip R-CNN-S achieves an improvement of 1.67% while using 0.4% fewer parameters and only 74.3% of the computations required by PKINet-S [1]. Additionally, Strip R-CNN-S shows a 2.57% enhancement over LSKNet-S [32] utilizing 1.1% fewer parameters and 2.2% less computations.

Results on DOTA-v1.0 [62]. We conduct a comparative analysis of different models and present detailed results for mean Average Precision (mAP) and Average Precision (AP) across categories on the DOTA dataset(refer to more model results in Supplementary Materials). As shown in Tab. 3, our single-scale evaluation demonstrates a 1.67% improve-

Table 3. Comparisons with SOTA methods on the DOTA-v1.0 dataset with single-scale and multi-scale training and testing. The StripNet-S backbone is pretrained on ImageNet for 300 epochs. †: Model ensemble as in MoCAE [49].

Method	**Pre.**	**mAP ↑**	**#P ↓**	**FLOPs ↓**	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Single-Scale																			
EMO2-DETR [27]	IN	70.91	74.3M	304G	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04
CenterMap [60]	IN	71.59	41.1M	198G	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59
AO2-DETR [8]	IN	72.15	74.3M	304G	86.01	75.92	46.02	66.65	79.70	79.93	89.17	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96
SASM [23]	IN	74.92	36.6M	-	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37
O-RCNN [64]	IN	75.87	41.1M	199G	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28
R3Det-GWD [71]	IN	76.34	41.9M	336G	88.82	82.94	55.63	72.75	78.52	83.10	87.46	90.21	86.36	85.44	64.70	61.41	73.46	76.94	57.38
COBB [63]	IN	76.52	41.9M	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R3Det-KLD [72]	IN	77.36	41.9M	336G	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	78.49	70.89
ARC [73]	IN	77.35	74.4M	217G	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43
LSKNet-S [32]	IN	77.49	31.0M	161G	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82
PKINet-S [1]	IN	78.39	30.8M	190G	89.72	84.20	55.81	77.63	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94
RTMDet-R [43]	IN	78.85	52.3M	205G	89.43	84.21	55.20	75.06	80.81	84.53	88.97	90.90	87.38	87.25	63.09	67.87	78.09	80.78	69.13
⋆ Strip R-CNN-S	IN	**80.06**	**30.5M**	**159G**	88.91	86.38	57.44	76.37	79.73	84.38	88.25	90.86	86.71	87.45	69.89	66.82	79.25	82.91	75.58
Multi-Scale																			
CSL [67]	IN	76.17	37.4M	236G	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
DODet [6]	IN	80.62	-	-	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47
AOPG [5]	IN	80.66	-	-	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10
KFIoU [73]	IN	80.93	58.8M	206G	89.44	84.41	62.22	82.51	80.10	86.07	88.68	90.90	87.32	88.38	72.80	71.95	78.96	74.95	75.27
PKINet-S [1]	IN	81.06	30.8M	190G	89.02	86.73	58.95	81.20	80.41	84.94	88.10	90.88	86.60	87.28	67.10	74.81	78.18	81.91	70.62
RTMDet-R [43]	CO	81.33	52.3M	205G	88.01	86.17	58.54	82.44	81.30	84.82	88.71	90.89	88.77	87.37	71.96	71.18	81.23	81.40	77.13
RVSA [58]	MA	81.24	114.4M	414G	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	73.11	84.75	81.88	77.58
LSKNet-S [32]	IN	81.64	31.0M	161G	89.57	86.34	63.13	83.67	82.20	86.10	88.66	90.89	88.41	87.42	71.72	69.58	78.88	81.77	76.52
⋆ Strip R-CNN-T	IN	81.40	20.5M	123G	89.14	84.90	61.78	83.50	81.54	85.87	88.64	90.89	88.02	87.31	71.55	70.74	78.66	79.81	78.16
⋆ Strip R-CNN-S	IN	**82.28**	**30.5M**	**159G**	89.17	85.57	62.40	83.71	81.93	86.58	88.84	90.86	87.97	87.91	72.07	71.88	79.25	82.45	82.82
⋆ Strip R-CNN-S†	IN	**82.75**	30.5M	159G	88.99	86.56	61.35	83.94	81.70	85.16	88.57	90.88	88.62	87.36	75.13	74.34	84.58	81.49	82.56

Table 4. Comparisons with SOTA methods on the DOTA-v1.5 dataset with single-scale training and testing. The StripNet-S backbone is pretrained on ImageNet for 300 epochs.

Method	RetinaNet-O [36]	FR-O [53]	Mask RCNN [22]	HTC [4]	ReDet [20]	DCFL [65]	LSKNet-S [32]	PKINet-S [1]	**Strip R-CNN-S**
mAP (%)	59.16	62.00	62.67	63.40	66.86	67.37	70.26	71.47	**72.27**

ment over PKINet-S. Furthermore, with multi-scale training and testing, we achieve 82.28% mAP for a single model. By ensembling the results of RTMDet and Strip R-CNN, following the model ensemble strategy in MoCAE [49], we achieve 82.75% mAP, setting a new state-of-the-art record.

Results on DOTA-v1.5 [62]. In this dataset with minuscule instances, as shown in Tab. 4, our approach achieves outstanding performance, demonstrating its efficacy and generalization ability to small objects. Our Strip R-CNN outperforms the former state-of-the-art method, achieving an improvement of 0.8%.

Results on FAIR1M-v1.0 [56]. The results in Tab. 5 reveal that our Strip R-CNN reaches a highly competitive mAP score of 48.26%. Our method could improve 0.39% mAP for LSKNet [32] and surpassing all other methods by a significant margin.

Results on DIOR-R [5]. As shown in Tab. 7, we observe 2.80% improvement over LSKNet [32] and 1.67% improvement over PKINet [1].

Results on HRSC2016 [38]. We achieve 98.70% mAP under the VOC2012 [14] metric, which is the state-of-the-art performance with 0.16% mAP improvement over PKINet [1] and 0.24% mAP over LSKNet [32]. The results are shown in Tab. 6.

Across multiple datasets, our method consistently surpasses previous state-of-the-art approaches, demonstrating its generalizability and effectiveness.

4.3. Ablation Studies

Kernel size of strip convolutions. The kernel size in strip convolutions is critical for our proposed strip module. We experiment with large kernel sizes, starting from 11 and increasing in increments of 4. Smaller kernels are ineffective in capturing features of high aspect ratio objects, while a kernel size of 15 yielded satisfactory results, with further improvement observed at 19. To further show the advantage of large strip convolution, we add a visual analysis of the feature representations of different kernel sizes. Fig. 8 shows that strip convs with a larger kernel size can learn the

Table 5. Comparisons with SOTA methods on the FAIR1M-v1.0 dataset. The StripNet-S backbone is pretrained on ImageNet for 300 epochs. *: Results are referenced from the FAIR1M paper [56].

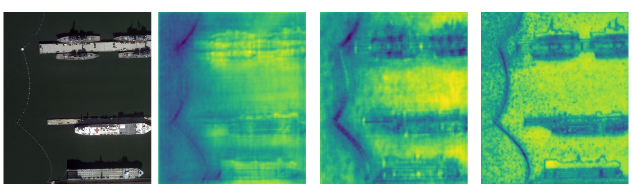
Method	G. V.* [66]	RetinaNet* [36]	C-RCNN* [2]	F-RCNN* [53]	RoI Trans.* [10]	O-RCNN [64]	LSKNet-S [32]	**Strip R-CNN-S**
mAP (%)	29.92	30.67	31.18	32.12	35.29	45.60	47.87	**48.26**

Table 6. Comparisons with SOTA methods on HRSC2016. mAP (07/12): VOC 2007 [13]/2012 [14] metrics. The StripNet-S backbone is pretrained on ImageNet for 300 epochs.

Method	Pre.	mAP(07)↑	mAP(12)↑	#P ↓	FLOPs ↓
DRN [50]	IN	-	92.70	-	-
DAL [45]	IN	89.77	-	36.4M	216G
GWD [71]	IN	89.85	97.37	47.4M	456G
AOPG [5]	IN	90.34	96.22	-	-
O-RCNN [64]	IN	90.50	97.60	41.1M	199G
RTMDet [43]	CO	90.60	97.10	52.3M	205G
LSKNet-S [32]	IN	90.65	98.46	31.0M	161G
PKINet-S [1]	IN	90.65	98.54	30.8M	190G
⋆ Strip R-CNN-S	IN	90.60	**98.70**	30.5M	159G

Table 7. Comparisons with SOTA methods on DIOR-R. The StripNet-S backbone is pretrained on ImageNet for 300 epochs.

Method	Pre.	#P ↓	FLOPs ↓	mAP (%)
RetinaNet-O [36]	IN	-	-	57.55
Faster RCNN-O [53]	IN	41.1M	198G	59.54
TIOE-Det [47]	IN	41.1M	198G	61.98
O-RCNN [64]	IN	41.1M	199G	64.30
ARS-DETR [79]	IN	41.1M	198G	66.12
O-RepPoints [29]	IN	36.6M	-	66.71
DCFL [65]	IN	-	-	66.80
LSKNet-S [32]	IN	31M	161G	65.90
PKINet-S [1]	IN	30.8M	190G	67.03
⋆ Strip R-CNN-S	IN	30.5M	159G	**68.70**



Input Image Kernel size: 11 Kernel size: 15 Kernel size: 19

Figure 8. Feature map visulization of different kernel sizes. Kernel size 19 leads to more accurate features of high aspect ratio objects.

features of slender objects with more precise localization information and well sharpen the object boundaries. We also test various kernel sizes at four stages in our StripNet backbone, exploring both increasing and decreasing strategies. Larger kernels in shallow layers combined with smaller kernels in deeper layers produce good results. In contrast, using smaller kernels in the shallow layers leads to significant performance drops, indicating that shallow layers benefit from larger receptive fields. Consequently, we select 19 as the optimal kernel size for the strip module at all stages of StripNet backbone.

Table 8. Ablation study on the kernel size of our proposed strip module at four stages of the StripNet backbone network. We adopt the StripNet-S backbone pretrained on ImageNet for 100 epochs. The best result is obtained when using kernel size 19 at all stages.

Kernel Size	#P↓	FLOPs↓	mAP (%)
(19,19,19,19)	13.30M	52.34G	**81.75**
(15,15,15,15)	13.28M	52.19G	81.64
(11,11,11,11)	13.26M	52.03G	81.22
(15,17,19,21)	13.31M	52.26G	81.37
(21,19,17,15)	13.29M	52.34G	81.72

Table 9. Ablation study on the design of our proposed strip module. We adopt the StripNet-S backbone pretrained on ImageNet for 100 epochs. d: dilation rate.

5×5 Square Conv	Large Strip Conv		#P ↓	FLOPs ↓	mAP (%)
	Sequential	Parallel			
✗	✓	✗	13.23M	51.84G	81.38
✓	✓	✗	13.30M	52.34G	**81.75**
✓	✗	✓	13.33M	52.52G	81.54
✓	19×19 Square Conv		14.17M	58.41G	81.44
✓	7×7 Square Conv d=3		13.30M	52.34G	81.55

Ablation on strip module design. We perform ablation experiments to analyze the design choices in the strip module as shown in Tab. 9. First, we assess the role of depth-wise square convolution. Removing this component leads to a large performance drop, emphasizing the importance of square convolution for capturing features of square-shaped objects. Next, we examine the integration of horizontal and vertical large strip convolutions, comparing parallel and sequential arrangements. The sequential configuration outperforms the parallel one, as the latter lacks effective two-dimensional modeling, merely combining the two strip convolutions without capturing the overall object structure. Furthermore, substituting the sequential large strip convolutions with either a 19×19 large kernel convolution or a 7×7 dilated convolution with a dilation rate of 3 results in noticeable performance loss, further validating the effectiveness of the large strip convolutions.

Effectiveness of the StripNet backbone. To validate the generality and effectiveness of our proposed StripNet backbone, we conduct experiments using various remote sensing detection frameworks. As can be seen in Tab. 17, Our method improves RoI Transformer [10] by 7.11%. It also

Table 10. Effectiveness of StripNet-S backbone on other remote sensing object detection frameworks. The StripNet-S backbone is pretrained on ImageNet [9] for 100 epochs.

Method	**Backbone**	**#P↓**	**FLOPs↓**	**mAP** (%)
RoI Trans. [10]	ResNet-50 [21]	41.1M	211.4G	74.61
	StripNet-S	**13.3M**	**52.3G**	**81.72**
S^2ANet [19]	ResNet-50 [21]	41.1M	211.4G	79.42
	StripNet-S	**13.3M**	**52.3G**	**80.52**
R3Det [68]	ResNet-50 [21]	41.1M	211.4G	76.47
	StripNet-S	**13.3M**	**52.3G**	**79.01**
YOLO [43]	CSPNeXt-l [21]	52.3M	80.2G	78.02
	StripNet-S	**13.3M**	**52.3G**	**78.50**
ARS-DETR [79]	ResNet-50 [21]	41.1M	211.4G	72.08
	StripNet-S	**13.3M**	**52.3G**	**75.07**

enhances S^2ANet and R3Det by 0.9% and 2.54%, respectively. When integrated into YOLO and DETR frameworks, our approach consistently boosts performance, demonstrating its broad compatibility and effectiveness in enhancing various detectors.

Ablation on strip head design. In designing the strip head, we initially decouple the head into (x, y), (w, h), and θ branches, applying the strip module to either the (x, y) or (w, h) branches. This approach shows moderate improvement. Combining the (x, y) and (w, h) branches yields slightly better results, probably due to the complementary and similar features shared by object position and shape information. For the θ branch, we found that fully connected layers slightly outperform convolutional layers. Therefore, we adopt two fully connected layers for the θ branch, and share them with the classification branch in that the sensitive regions for classification and angle predictions share some overlap and exhibit some complementary characteristics, as discussed in Sec. 3.3. Simply applying the strip module to the localization branch produces noticeable improvements. We argue that more specialized structures tailored to different regression parameters could further enhance performance. Results are shown in Tab. 11.

Effectiveness of the strip head. As can be seen in Tab. 12, we compare the performance of two different detectors before and after incorporating the strip head. The results demonstrate that our strip head consistently improves the performance of other detectors, confirming its effectiveness and generalizability. For the ROI Transformer, our method achieves a significant improvement of 5.0 mAP, which is a remarkable increase. Similarly, the other detector Rotated Faster R-CNN also receives performance gains, further validating that our approach can be used for a wide range of detectors and is able to enhance their detection performance.

Table 11. Ablation study of the **strip head design**. *fc* refers to using two fully connected layers. *conv* refers to using two 3×3 convolutional layers. *strip* refers to replacing the latter 3×3 convolutional layer with our designed strip module. We adopt the StripNet-S backbone pretrained on ImageNet for 100 epochs.

Different Head Structure	**mAP** (%)
$(x, y, w, h, cls, \theta)_{fc}$	81.75
$(x, y)_{conv}, (w, h)_{conv}, (cls)_{fc}, (\theta)_{fc}$	81.76
$(x, y)_{conv}, (w, h)_{conv}, (cls)_{fc}, (\theta)_{conv}$	81.72
$(x, y)_{strip}, (w, h)_{conv}, (cls)_{fc}, (\theta)_{fc}$	81.88
$(x, y)_{conv}, (w, h)_{strip}, (cls)_{fc}, (\theta)_{fc}$	81.77
$(x, y)_{conv}, (w, h)_{conv}, (cls)_{fc}, (\theta)_{strip}$	81.81
$(x, y, w, h)_{strip}, (cls)_{fc}, (\theta)_{fc}$	81.86
$(x, y, w, h)_{strip}, (cls, \theta)_{fc}$	**81.96**
$(x, y, w, h)_{strip}, (cls)_{fc}, (\theta)_{strip}$	81.93

Table 12. Effectiveness of strip head on other rotated detectors.

Method	**Head**	**mAP** (%)
Faster RCNN-O [53]	Original Head	76.17
	Strip Head	77.88
RoI Trans. [10]	Original Head	74.61
	Strip Head	79.37

4.4. Visual Analysis

We also present detection results and Eigen-CAM [48] visualizations in Fig. 9. We can see that for high aspect ratio objects, previous methods such as Oriented-RCNN and LSKNet have issues like missed detections and significant localization errors. In contrast, our method can successfully detect the high aspect ratio objects. The Eigen-CAM visualizations also show strong activations of our method for these objects, validating the effectiveness of our approach.

Additionally, to further substantiate the superiority of our method in detecting high aspect ratio objects, we conduct additional experiments on the DOTA dataset. The models are trained on the DOTA train set and tested on the val set, following the evaluation method used in [80]. We assess detection performance across objects with different aspect ratios. As shown in Fig. 1, the results indicate that within the aspect ratio range of 1-5, our method and previous approaches show comparable performance, with only minor differences. However, for objects with larger aspect ratios, our method demonstrates an obvious advantage.

5. Conclusions

In this paper, we alleviate the challenge of detecting slender objects in remote sensing scenarios by leveraging large strip convolutions to better extract features and improve localization of such objects. Based on large strip convolutions, we propose the simple yet highly effective Strip R-CNN. Extensive experiments demonstrate that our method exhibits strong generalization capability and achieves state-of-the-art performance on several remote sensing benchmarks. We

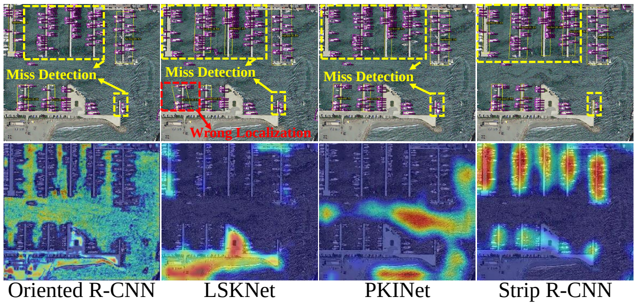


Figure 9. **Top:** Detection results. Our method Strip R-CNN can successfully capture the high aspect ratio objects. **Bottom:** Eigen-CAM visualizations. The Eigen-CAM visualization of our method shows strong activations for high aspect ratio objects, further validating the effectiveness of our approach.

hope this research could facilitate the development of object detection in the remote sensing field.

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Strip R-CNN: Large Strip Convolution for Remote Sensing Object Detection

Supplementary Material

6. Dataset details

DOTA-v1.0 [62] is a large-scale dataset for remote sensing detection which contains 2806 images, 188,282 instances, and 15 categories: Plane (PL), Baseball diamond (BD), Bridge (BR), Ground track field (GTF), Small vehicle (SV), Large vehicle (LV), Ship (SH), Tennis court (TC), Basketball court (BC), Storage tank (ST), Soccer-ball field (SBF), Roundabout (RA), Harbor (HA), Swimming pool (SP), and Helicopter (HC). Eeah image is of the size in the range from 800×800 to 20000×20000 pixels and contains objects exhibiting a wide variety of scales, orientations, and shapes. The proportions of the training set, validation set, and testing set in DOTA-v1.0 are 1/2, 1/6, and 1/3, respectively. For single scale training and testing, we first crop the image into 1024×1024 patches with an overlap of 200 following the previous methods [1, 32, 51],, we use both the training set and validation set for training, and the testing set for test.

DOTA-v1.5 [62] is a more challenging dataset using the same images as DOTA-v1.0, but the extremely small instances (less than 10 pixels) are also annotated. In this version, a new category, "Container Crane" (CC), has been introduced, and the number of small instances has significantly increased, with a total of 403,318 instances across the dataset. The number of images and dataset splits remains unchanged from DOTA-v1.0.

FAIR1M-v1.0 [56] is a recently published remote sensing dataset that consists of 15,266 high-resolution images and more than 1 million instances. All objects in the FAIR1M dataset are annotated with respect to 5 categories and 37 sub-categories by oriented bounding boxes. Each image is of the size in the range from 1000×1000 to $10,000 \times 10,000$ pixels and contains objects exhibiting a wide variety of scales, orientations, and shapes. The number of images of the train set and testing set in FAIR1M-1.0 are 16,488 and 8,137, respectively. During training, we apply the same preprocessing methods as DOTA-v1.0 dataset, resulting in multi-scale training and testing sets.

HRSC2016 [38] is a remote sensing dataset for ship detection that contains 1061 aerial images whose size ranges from 300×300 and 1500×900. All images are sourced from Google Earth, and the dataset is divided into training, validation, and test sets, with 436 images for training, 181 for validation, and 444 for testing.

DIOR-R [5] dataset extends the DIOR remote sensing dataset by providing oriented bounding box (OBB) annotations. It includes 23,463 images, each with a resolution of 800×800 pixels, and a total of 192,518 annotations.

Table 13. Comparisons with large-kernel-based methods on the DOTA-v1.0 [62] dataset. Params and FLOPs are computed for backbone only. The StripNet-S backbone is pretrained on ImageNet for 300 epochs.

Method	**Pre.**	**#P ↓**	**FLOPs ↓**	**mAP (%)**
VAN-B1 [16]	IN	13.4M	52.7G	81.15
SegNext [17]	IN	13.1M	45.0G	81.12
ConvNext V2-N [40]	IN	15.0M	51.2G	80.81
⋆ Strip R-CNN-S	IN	13.3M	52.3G	**82.28**

7. Implementation details

In this paper, we adopt the approach outlined in previous works [1, 32] to implement multi-scale training and testing across the DOTA, DOTAv1.5, and FAIR1M datasets. Specifically, we rescale each image to three different scales (0.5, 1.0, 1.5) and then crop them into 1024×1024 patches, with a 500-pixel overlap between adjacent patches. It is important to note that throughout both the pre-training and fine-tuning stages, we did not utilize Exponential Moving Average (EMA) techniques. As a result, the performance metrics reported for LSKNet in the main table are based on results obtained without EMA.

8. FAIR1M benchmark results

Fine-grained category result comparisons with state-of-the-art methods on the FAIR1M-v1.0 dataset are given in Tab. 14

9. DOTA benchmark results

Fine-grained category result comparisons with state-of-the-art methods on the DOTA-v1.0 and DOTA-v1.5 dataset are given in Tab. 16 and Tab. **??**

10. More results of the effectiveness of the strip head

More result comparisons of two different detectors before and after incorporating the strip head are shown in Tab. 18. The results is obtained on DOTA-v1.0 dataset with single-scale training and testing. As can be seen in Tab. 18, we compare the performance of two different detectors before and after incorporating the strip head. The results demonstrate that our strip head consistently improves the performance of other detectors, confirming its effectiveness and generalizability. For LSKNet [32], our method achieves a

Table 14. Comparisons With the SOTA Models on the FAIR1M-v1.0 Datasets. *The correspondence between C1-C34 and object categories in the table (in order). C1: Boeing 737, C2: Boeing 747, C3: Boeing 777, C4: Boeing 787, C5: C919, C6: A220, C7: A321, C8: A330, C9: A350, C10: ARJ21, C11: passenger ship, C12: motorboat, C13: fishing boat, C14: tugboat, C15: engineering ship, C16: liquid cargo ship, C17: dry cargo ship, C18: warship, C19: small car, C20: bus, C21: cargo truck, C22: dump truck, C23: van, C24: trailer, C25: tractor, C26: excavator, C27: truck tractor, C28: basketball court, C29: tennis court, C30: football field, C31: baseball field, C32: intersection, C33: roundabout, C34: bridge.*

Method	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**	**C8**	**C9**	**C10**	**C11**	**C12**	**C13**	**C14**	**C15**	**C16**	**C17**	**mAP** (%)
	C18	**C19**	**C20**	**C21**	**C22**	**C23**	**C24**	**C25**	**C26**	**C27**	**C28**	**C29**	**C30**	**C31**	**C32**	**C33**	**C34**	
Gliding Vertex [66]	35.43	47.88	15.67	48.32	0.01	40.11	39.31	16.54	16.56	0.01	9.12	23.34	1.23	15.67	15.43	15.32	25.43	29.92
	13.56	66.23	23.43	46.78	36.56	53.78	14.32	16.39	16.92	28.91	48.41	80.31	53.46	66.93	59.41	16.25	10.39	
RetinaNet [36]	38.46	55.36	24.57	51.84	0.81	40.5	41.06	18.02	19.94	1.7	9.57	22.54	1.33	16.37	19.11	14.26	24.7	30.67
	15.37	65.2	22.42	44.17	35.57	52.44	19.17	1.28	17.03	28.98	50.58	81.09	52.5	66.76	60.13	17.41	12.58	
Cascade-RCNN [2]	40.42	52.86	29.07	52.47	0	44.37	38.35	26.55	17.54	0	12.1	28.84	0.71	15.35	18.53	14.63	25.15	31.18
	14.53	68.19	28.25	48.62	40.4	58	13.66	0.91	16.45	30.27	38.81	80.29	48.21	67.9	55.67	20.35	12.62	
Faster-RCNN [53]	36.43	50.68	22.5	51.86	0.01	47.81	43.83	17.66	19.95	0.13	9.81	28.78	1.77	17.65	16.47	16.19	27.06	32.12
	13.16	68.42	28.37	51.24	43.6	57.51	15.03	3.04	17.99	29.36	58.26	82.67	54.5	71.71	59.86	16.92	11.87	
RoI Transformer [10]	39.58	73.56	18.32	56.43	0	47.67	49.91	27.64	31.79	0	14.31	28.07	1.03	14.32	15.97	18.04	26.02	35.29
	12.97	68.8	37.41	53.96	45.68	58.39	16.22	5.13	22.17	46.71	54.84	80.35	56.68	69.07	58.44	18.58	31.81	
OFA-Net [46]	43.20	87.58	23.58	57.78	30.71	54.31	67.83	73.65	73.56	29.25	9.90	68.15	11.70	30.37	11.83	26.80	38.68	43.73
	35.42	69.62	33.67	44.69	49.23	71.61	13.45	6.06	17.36	5.35	49.20	83.75	61.29	88.77	57.97	24.68	35.82	
O-RCNN [64]	38.65	84.96	20.05	55.57	25.55	48.58	68.75	71.18	74.43	32.85	20.46	67.63	11.13	32.89	16.22	26.45	39.03	44.30
	34.27	70.88	45.01	48.98	52.45	70.19	14.50	4.42	19.81	7.51	56.30	80.41	68.13	89.06	57.92	21.87	30.11	
CHODNet [54]	40.23	53.39	29.65	54.04	0.02	46.26	43.12	27.61	20.89	0.01	12.34	29.34	1.71	17.77	17.72	16.78	27.51	32.46
	13.68	70.12	28.38	49.11	44.02	60.78	14.48	4.96	17.60	30.09	47.91	82.11	54.10	69.97	19.57	59.91	14.38	
OBB-ISP [55]	43.97	88.77	38.97	58.29	50.88	54.09	65.14	59.19	65.94	35.00	19.78	56.77	26.84	26.31	21.41	44.97	47.39	45.91
	34.30	61.16	34.10	50.04	52.31	56.24	15.99	7.43	17.03	37.14	54.99	81.43	53.56	87.66	54.25	24.20	35.46	
LSKNet-S [32]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	47.87
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Ours																		
$\star$Strip R-CNN-S	42.17	89.07	21.52	55.24	23.03	50.97	73.58	72.96	78.29	40.81	20.78	71.52	16.12	42.86	17.70	28.55	40.11	**48.26**
	39.56	76.14	49.44	55.37	60.55	76.18	20.32	8.12	25.96	7.23	58.94	86.59	72.73	89.43	62.89	25.57	40.58	

improvement of 0.56%. Similarly, Oriented R-CNN [64] also receives a 0.3% performance gain, further validating that our approach is compatible with a wide range of detectors and is able to reliably enhance detection performance.

11. The effectiveness of the StripNet backbone

To validate the generality and effectiveness of our proposed StripNet backbone, we conduct experiments using various remote sensing detection frameworks. These include two-stage frameworks such as O-RCNN [64] and RoI Transformer [10], as well as one-stage frameworks like S^2ANet [19] and R3Det [68]. Result comparisons of two different detectors before and after incorporating the StripNet-S are shown in Tab. 17. The results is obtained on DOTA-v1.0 dataset with multi-scale training and testing. As can be seen in Tab. 17, we compare the performance of two different detectors before and after incorporating the StripNet-S backbone. The results demonstrate that our StripNet-S backbone consistently improves the performance of other detectors, confirming its effectiveness and generalizability. For RoI Transformer [10], our method achieves a improvement of 7.11%. Similarly, Oriented R-CNN [64] also receives a 0.88% performance gain, S^2ANet and R3Det obtains a improvement of 0.9% and 2.54% respectively, further validating that our approach is compatible with a wide range of detectors and is able to reliably enhance detection performance.

12. Comparisons with other large kernel networks

We also compare our Strip R-CNN-S with some popular high-performance backbone models which using large kernels inTab. 13. In the three models under discussion, all have adopted large convolutional kernels. Specifically, VAN [16] utilizes dilated convolutions to expand the receptive field. While this technique effectively increases the model's receptive feild, it can also lead to the neglect of certain critical information, which is particularly disadvantageous for densely distributed objects in remote sensing scenes. SegNext [17] introduces a multi-branch large convolutional kernel structure, designed to capture richer feature information through multiple pathways. However, this

Table 15. Comparisons with SOTA methods on the DOTA-v1.0 dataset with single-scale and multi-scale training and testing. The StripNet-S backbone is pretrained on ImageNet for 300 epochs. †: Model ensemble as in MoCAE [49].

Method	Pre.	mAP↑	#P↓	FLOPs↓	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Single-Scale																			
EMO2-DETR [27]	IN	70.91	74.3M	304G	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04
CenterMap [60]	IN	71.59	41.1M	198G	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59
AO2-DETR [8]	IN	72.15	74.3M	304G	86.01	75.92	46.02	66.65	79.70	79.93	89.17	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96
SCRDet [69]	IN	72.61	41.9M	-	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21
R3Det [68]	IN	73.70	41.9M	336G	89.5	81.2	50.5	66.1	70.9	78.7	78.2	90.8	85.3	84.2	61.8	63.8	68.2	69.8	67.2
RoI Trans. [10]	IN	74.05	55.1M	200G	89.01	77.48	51.64	72.07	74.43	77.55	87.76	90.81	79.71	85.27	58.36	64.11	76.50	71.99	54.06
S^2ANet [19]	IN	74.12	38.5M	-	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94
SASM [23]	IN	74.92	36.6M	-	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37
G.V. [66]	IN	75.02	41.1M	198G	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32
O-RCNN [64]	IN	75.87	41.1M	199G	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28
ReDet [20]	IN	76.25	31.6M	-	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59
R3Det-GWD [71]	IN	76.34	41.9M	336G	88.82	82.94	55.63	72.75	78.52	83.10	87.46	90.21	86.36	85.44	64.70	61.41	73.46	76.94	57.38
COBB [63]	IN	76.52	41.9M	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R3Det-KLD [72]	IN	77.36	41.9M	336G	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	78.49	70.89
ARC [73]	IN	77.35	74.4M	217G	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43
LSKNet-S [32]	IN	77.49	31.0M	161G	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82
PKINet-S [1]	IN	78.39	30.8M	190G	89.72	84.20	55.81	77.63	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	62.94
RTMDet-R [43]	IN	78.85	52.3M	205G	89.43	84.21	55.20	75.06	80.81	84.53	88.97	90.90	87.38	87.25	63.09	67.87	78.09	80.78	69.13
⋆ Strip R-CNN-S	IN	**80.06**	**30.5M**	**159G**	88.91	86.38	57.44	76.37	79.73	84.38	88.25	90.86	86.71	87.45	69.89	66.82	79.25	82.91	75.58
Multi-Scale																			
G.V. [66]	IN	75.02	41.1M	198G	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32
R3Det [68]	IN	76.47	41.9M	336G	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.57	62.68	67.53	78.56	72.62
CSL [67]	IN	76.17	37.4M	236G	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
CFA [18]	IN	76.67	-	-	89.08	83.20	54.37	66.87	81.23	80.96	87.17	90.21	84.32	86.09	52.34	69.94	75.52	80.76	67.96
DAFNet [28]	IN	76.95	-	-	89.40	86.27	53.70	60.51	82.04	81.17	88.66	90.37	83.81	87.27	53.93	69.38	75.61	81.26	70.86
S^2ANet [19]	IN	79.42	-	-	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58
R3Det-GWD [71]	IN	80.23	41.9M	336G	89.66	84.99	59.26	82.19	78.97	84.83	87.70	90.21	86.54	86.85	73.47	67.77	76.92	79.22	74.92
DODet [6]	IN	80.62	-	-	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47
AOPG [5]	IN	80.66	-	-	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10
R3Det-KLD [72]	IN	80.63	41.9M	336G	89.92	85.13	59.19	81.33	78.82	84.38	87.50	89.80	87.33	87.00	72.57	71.35	77.12	79.34	78.68
KFIoU [73]	IN	80.93	58.8M	206G	89.44	84.41	62.22	82.51	80.10	86.07	88.68	90.90	87.32	88.38	72.80	71.95	78.96	74.95	75.27
PKINet-S [1]	IN	81.06	30.8M	190G	89.02	86.73	58.95	81.20	80.41	84.94	88.10	90.88	86.60	87.28	67.10	74.81	78.18	81.91	70.62
RTMDet-R [43]	CO	81.33	52.3M	205G	88.01	86.17	58.54	82.44	81.30	84.82	88.71	90.89	88.77	87.37	71.96	71.18	81.23	81.40	77.13
RVSA [58]	MA	81.24	114.4M	414G	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	73.11	84.75	81.88	77.58
LSKNet-S [32]	IN	81.64	31.0M	161G	89.57	86.34	63.13	83.67	82.20	86.10	88.66	90.89	88.41	87.42	71.72	69.58	78.88	81.77	76.52
⋆ Strip R-CNN-T	IN	81.40	20.5M	123G	89.14	84.90	61.78	83.50	81.54	85.87	88.64	90.89	88.02	87.31	71.55	70.74	78.66	79.81	78.16
⋆ Strip R-CNN-S	IN	**82.28**	**30.5M**	**159G**	89.17	85.57	62.40	83.71	81.93	86.58	88.84	90.86	87.97	87.91	72.07	71.88	79.25	82.45	82.82
⋆ Strip R-CNN-S†	IN	82.75	30.5M	159G	88.99	86.56	61.35	83.94	81.70	85.16	88.57	90.88	88.62	87.36	75.13	74.34	84.58	81.49	82.56

Table 16. Comparison with SOTA methods on the **DOTA-v1.5** dataset with single-scale training and testing.

Method	Pre.	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP(%)↑
RetinaNet-O [36]	IN	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR-O [53]	IN	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [22]	IN	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC [4]	IN	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
ReDet [20]	IN	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86
LSKNet-S [32]	IN	72.05	84.94	55.41	74.93	52.42	77.45	81.17	90.85	79.44	69.00	62.10	73.72	77.49	75.29	55.81	42.19	70.26
PKINet-S [1]	IN	80.31	85.00	55.61	74.38	52.41	76.85	88.38	90.87	79.04	68.78	67.47	72.45	76.24	74.53	64.07	37.13	71.47
⋆ Strip R-CNN-S	IN	80.04	83.26	54.40	75.38	52.46	81.44	88.53	90.83	84.80	69.65	65.93	73.28	74.61	74.04	69.70	38.98	**72.27**

architecture suffers from issues of feature confusion and feature redundancy, which may impact the model's performance. Although these three methods validate the effectiveness of large convolutional kernels, they fail to adequately address the unique characteristics of data specific to remote sensing scenarios. This limitation restricts their generaliza-

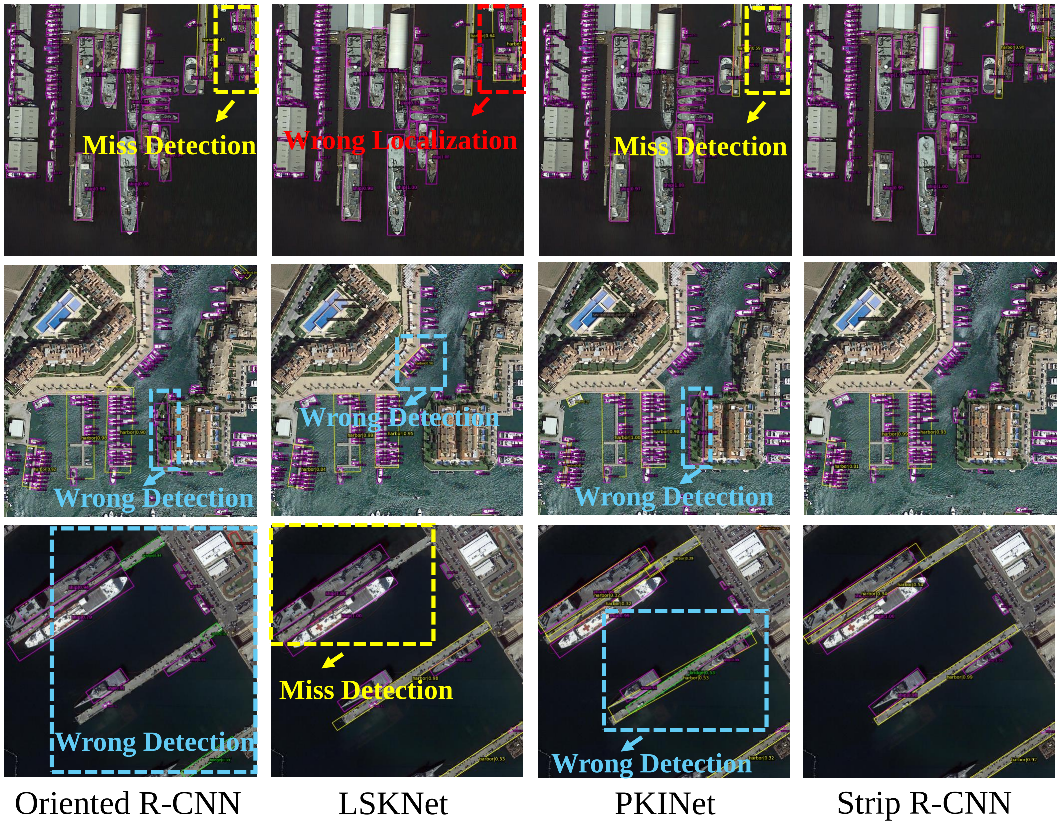

Figure 10. Detection results. Our method Strip R-CNN-S can successfully capture the high aspect ratio objects which validating the effectiveness of our approach.

Table 17. Effectiveness of StripNet-S backbone on other remote sensing object detection frameworks. The StripNet-S backbone is pre-trained on ImageNet [9] for 100 epochs.

Method	**Backbone**	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	**mAP (%)**
RoI Trans. [10]	ResNet-50 [21]	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
	StripNet-S	89.10	85.45	63.61	82.84	80.95	85.84	88.68	90.84	88.35	87.92	72.12	71.37	79.73	83.32	75.77	**81.72**
O-RCNN [64]	ResNet-50 [21]	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11	80.87
	StripNet-S	89.46	84.40	63.04	82.89	81.84	86.16	88.66	90.88	88.12	86.97	74.27	73.69	79.05	81.14	76.44	**81.75**
S^2ANet [19]	ResNet-50 [21]	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
	StripNet-S	89.31	84.94	61.01	81.87	81.34	85.10	88.69	90.76	86.89	87.49	67.94	68.74	77.96	78.82	73.90	**80.32**
R3Det [68]	ResNet-50 [21]	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.57	62.68	67.53	78.56	72.62	76.47
	StripNet-S	89.56	83.47	56.71	81.28	79.80	83.87	88.63	90.87	86.16	86.93	66.82	69.16	75.64	74.42	71.88	**79.01**

tion capabilities in such domains. In response to this challenge, our proposed method not only thoroughly analyzes the features of remote sensing data but also specifically optimizes for objects with high aspect ratios. Consequently, our solution demonstrates superior performance in remote sensing applications, offering new insights and directions for addressing challenges in this field.

13. More detection results

We present additional detection results in Fig. 10. As can be seen in Fig. 10, previous methods, including Oriented R-CNN, LSKNet, and PKINet, frequently struggle with detecting high aspect ratio objects, often resulting in missed or incorrect detections. In contrast, our approach, Strip R-CNN, effectively detects these challenging objects.

Table 18. Effectiveness of strip head on other remote sensing object detectors.

Method	**Head**	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	**mAP (%)**
LSKNet-S [32]	Original Head	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82	77.49
	Strip Head	89.71	83.50	56.18	78.86	79.83	84.66	88.16	90.86	88.30	86.11	67.02	66.27	76.19	71.79	63.39	**78.05**
O-RCNN [64]	Original Head	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.81
	Strip Head	89.48	82.33	53.95	72.83	78.60	77.97	88.12	90.89	85.84	84.76	64.42	67.55	74.57	70.31	60.05	**76.11**